

Design of IOT Architecture for Battery Charging Management System

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Abstract

Rapid growing of electric vehicle (EV) adoption faces challenges such as range anxiety and inconsistent charging infrastructure, which can be effectively addressed through an IoT-based architecture that offers real-time monitoring, predictive analytics, and mobile charging units, ultimately enhancing user experience, optimizing energy management, and promoting sustainable energy solutions for a cleaner future.

Introduction

•Growing Demand for Charging Infrastructure:

The rapid increase in electric vehicle (EV) sales is creating a pressing need for more efficient and accessible charging stations to alleviate range anxiety among users.

•Integration of Renewable Energy:

There is a critical requirement for sustainable charging solutions that incorporate renewable energy sources, such as solar power, to align with environmental sustainability goals.

•Cost Reduction and Innovation:

The necessity to reduce operational costs in charging management is driving the adoption of innovative IoT technologies aimed at optimizing battery management and enhancing overall efficiency.

Problem Statement

- Demand of Charging Infrastructure Expansion
- Grid Load Disturbances
- High Installation and Operational Costs
- Inefficient Load Management
- Limited Remote Monitoring

Objectives

- Develop an IoT-based architecture for optimized battery charging management.
- Incorporate real-time data monitoring and predictive analytics.
- Enhance operational cost efficiency by up to 88.5%.
- Promote sustainable energy integration through solar and ESS.

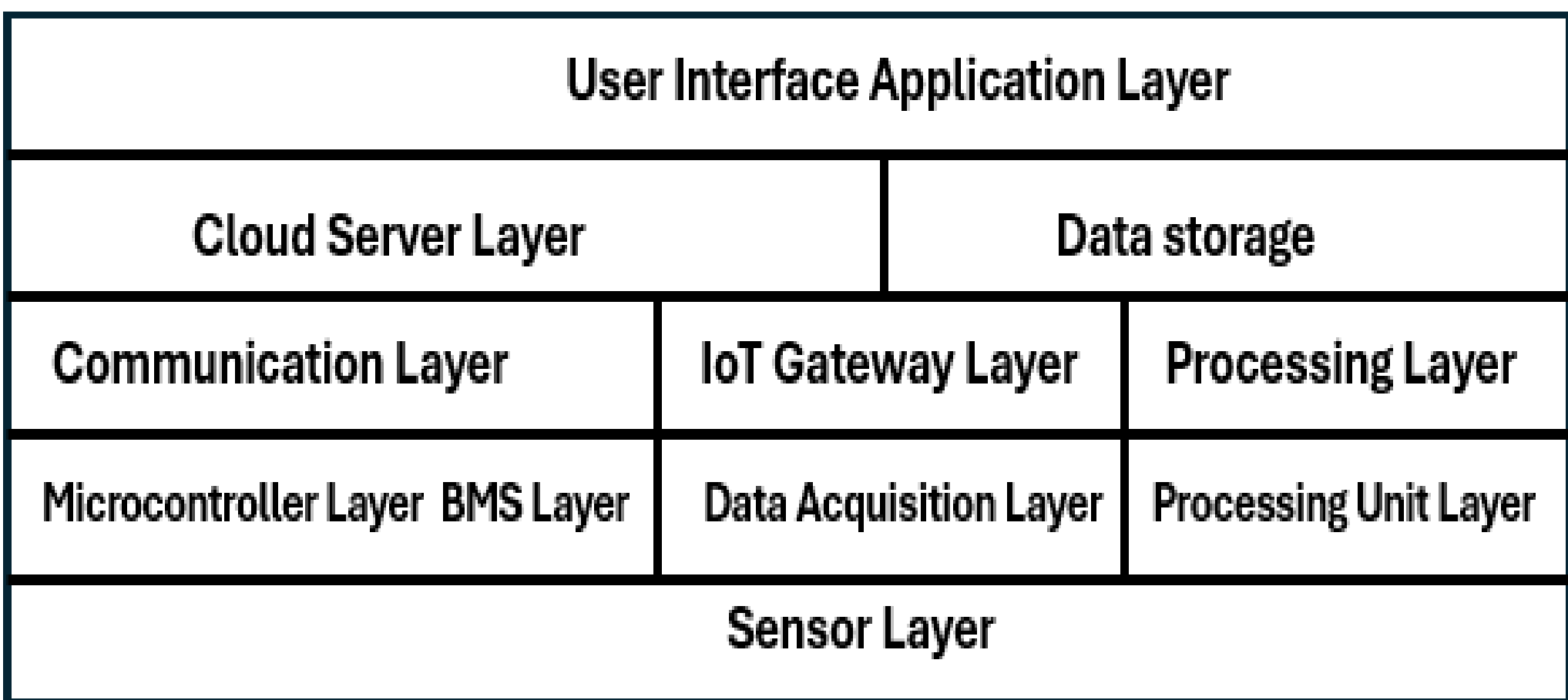
Project Novelty

- Innovative IOT 5 layered Architecture.
- Integration of multiple energy sources.
- Central monitoring and controlling.
- Real-Time data processing and user interface.
- Cloud connectivity.
- Cost efficiency model.
- Scalability and Adaptability.
- Portable charging station.

Methodology

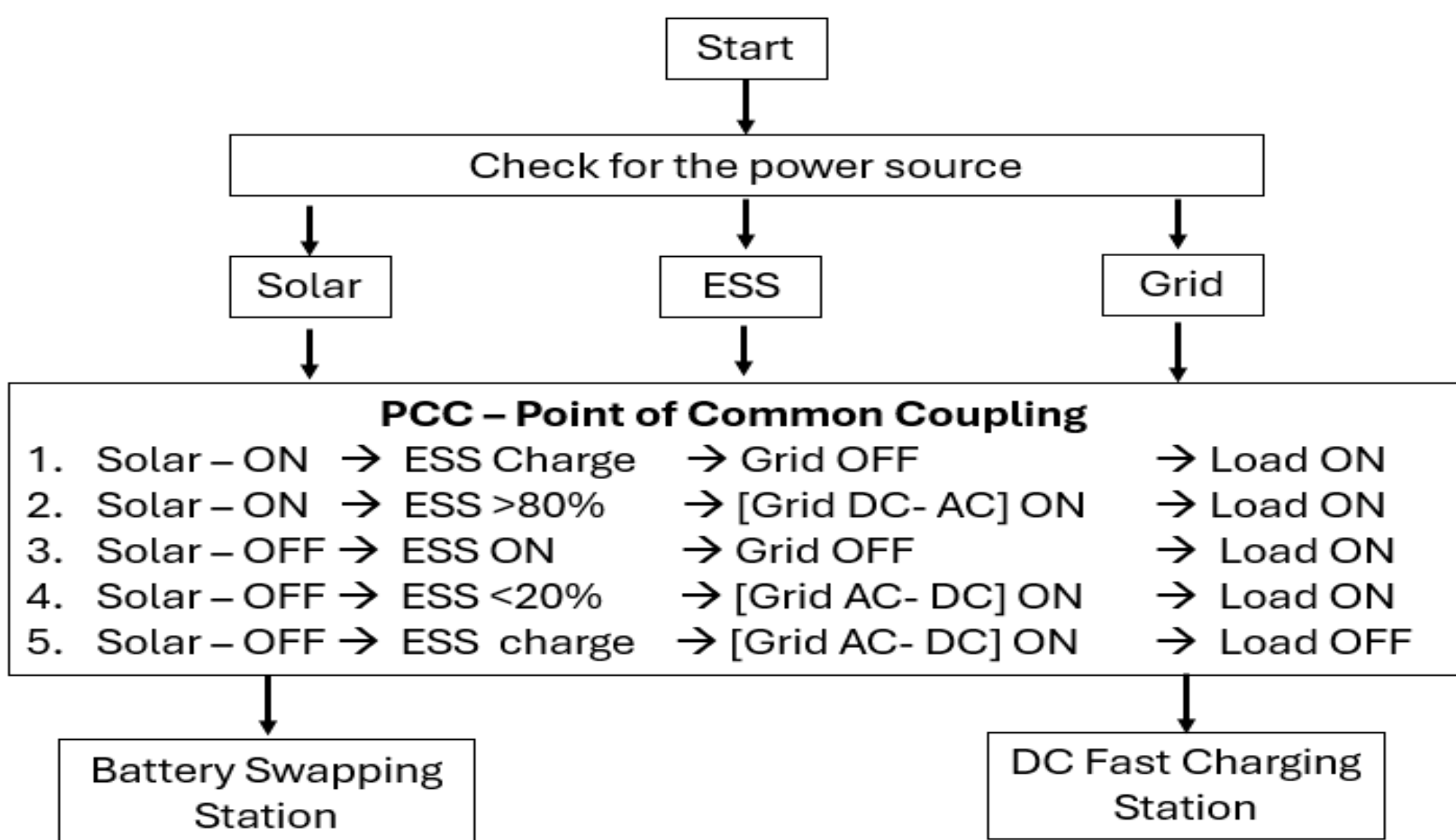
IoT Five-Layered Architecture

- User Interface Application Layer
- Cloud Server and Data Storage
- Communication Layer
- Microcontroller Layer
- Sensor Layer.



Point of Common Coupling (PCC)

Combines solar power, ESS, and grid power for



Algorithms

MPPT (Maximum Power Point Tracking) Algorithm:

Optimizes solar panel output by continuously adjusting the operating point to maximize energy capture under varying conditions.

Battery Swapping Station Algorithm:

Manages logistics and scheduling for battery swaps, optimizing inventory and availability of charged batteries to improve service efficiency.

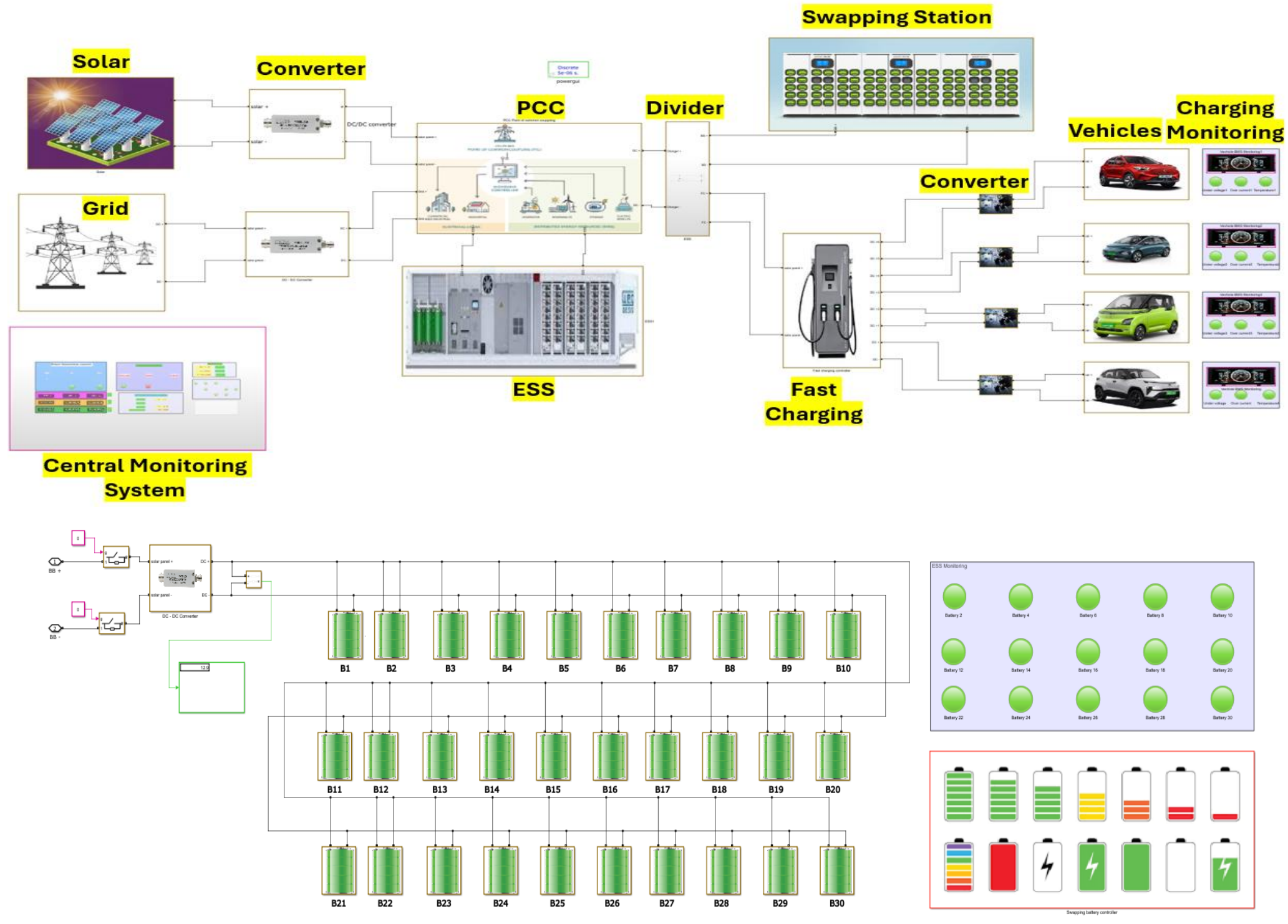
Vehicle Charging Control Calculation:

Calculates optimal charging rates based on battery state-of-charge (SOC) and user demand, ensuring safe and efficient charging.

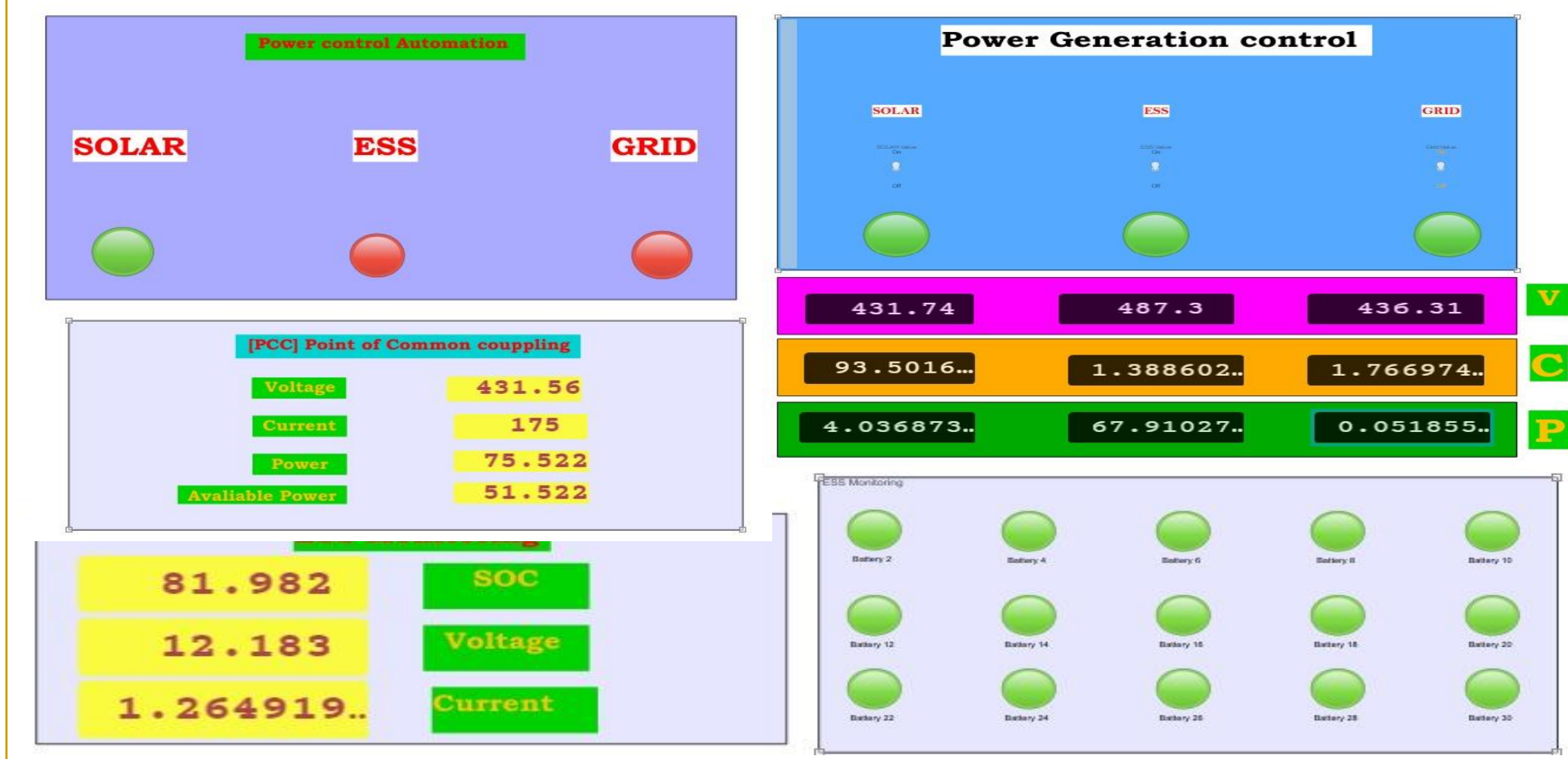
Safety and Blast Condition Prediction using ML:

Utilizes machine learning models to analyze historical data and predict potential safety hazards in battery charging environments, enabling proactive risk management.

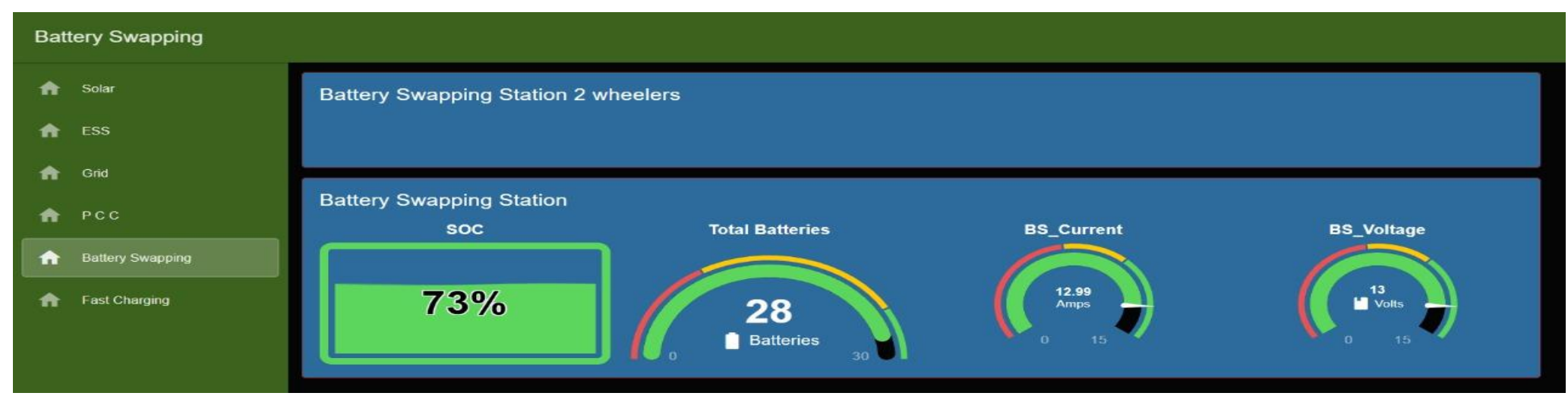
Simulation work



Real Time Monitoring



Data Visualization



Estimated Energy and Cost calculation

Power Generation							
Power Source	Voltage [V]	Current [I]	% of Generation/day	Required Power	Power [P] KW/hr	Power/day	
Solar	430	16.7	60	72	43	1032	
ESS	480	16.7	30	80	24	576	
Grid	435	20.0	10	87	9	209	
Total Power Generation per day in KW/24 hr						1817	
PCC - Point of common coupling			Battery Swapping Station				
PCC will Logically controlled on availability of source			Total Batteries				
			Under maintenance				
			Available batteries				
Voltage	12	Total battery charging		90			
Current	10	Cost of charge per battery		6			
Power	5	Total Amount Per Day		2700			
Power - [W] - KW/h	75	1800 Gen/day					
Power Availability	1.17	1772 use / day					
Fast Charging Station Load			ESS - Charging				
Cars	Voltage [V]	Power [kW]	Battery capacity [kWh]	SOC Available	Required current	Charging Time in (kWh) charge	No. of batteries
1-Tata Nano - EV	400	30.2	30.2	20	75.5	2.47	10
2-MG 25 EV	400	50	44.5	30	125	1.48	12
3-Tata Nano EV	400	16.8	26	50	42	4.40	100
4-Mahindra e20	400	70	60	10	175	1.07	48000
Total Power	167		Total Time	9	Total Revenue	2977	Power / one cycle charge - KW
Total Power	1670	KW power can charge / day	Total time	/ 56 hrs	Total Revenue	26794	48
Total Energy claculation							
Power sources	KW / day		KW / Month		KW / 5 years		
Required Full Power From Grid	1,817		54,510		663,205		
Solar	1,032	30,960	376,680	1,883,400			
ESS	576	17,280	210,240	1,051,200			
Grid	209	6,270	76,285	381,425			
Total Power saving / Day	1,608	48,240	586,920	2,934,600			
Total Generation Cost / Day 12hrs / unit	21,804	654,120	7,958,460	39,792,300			
Total Generation Cost Saving / Day 12hrs / unit	19,296	578,880	7,043,040	35,215,200			
Spending Amount	2,208	75,240	915,420	4,577,100			
				Revenue			
				Source	Per day	Per Month	
				Per Year	Per 5 Years		
Fast Charging Station				26,794	803,813	9,779,719	48,898,534
Battery Swapping				2,700	81,000	985,500	4,927,500
Total Revenue				29,494	884,813	10,765,219	53,826,034
Spending Amount				2,508	75,240	915,420	4,577,100
Total Profit				26,986	809,573	9,849,799	49,248,934

Conclusion

The proposed IoT-based architecture addresses critical challenges in EV charging infrastructure by enhancing efficiency, integrating renewable energy, and providing real-time monitoring.

This innovation paves the way for sustainable transportation and significant cost savings.

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References

- Cheng-Ta Chiang; Liang-Yu Chien "A Pitaya Dye-Sensitized Solar Cell Monitor for Environmental Sunlight Intensity Detection" IEEE Sensors Journal, 19-11, 03 February 2019.
- Saswat Kumar Ram; Sauvagya Ranjan Sahoo; Banee Bandana Das; Kamalakanta Mahapatra; Saraju P. Mohanty "Eternal-Thing: A Secure Aging-Aware Solar-Energy Harvester Thing for Sustainable IoT" IEEE Transactions on Sustainable Computing, 6-2, 15 April 2020.
- Tai-Yeon Ku; Wan-Ki Park; Hoon Choi "ESS Simulation based on PV Generation Prediction" 2018 International Conference on Information and Communication Technology Convergence (ICTC), 18 November 2018.
- Abdelrahman Ahmed; Sobhy Abdelkader; Ehab Ayman "IoT-based optimization of microgrid operation" 2023 24th International Middle East Power System Conference (MEPCON), 18 March 2024.
- George Y. Odongo; Richard Musabe; Damien Hanyurwimfura; Abubakar Diwani Bakari "An Efficient LoRa-Enabled Smart Fault Detection and Monitoring Platform for the Power Distribution System Using Self-Powered IoT Devices" IEEE Access (Volume: 10), 07 July 2022.
- Isa Hafidz; Ardono Priyadi; Margo Pujiatara; Dimas Okky Anggriawan; Mauridhi Hery Purnomo "Development of IoT-Based Portable Power Quality Monitoring on Microgrids by Enhancing Protection Features" IEEE Access (Volume: 11), 15 May 2023.
- Dan Zheng; Fushuan Wen; Jiansheng Huang "Optimal planning of battery swap stations" International Conference on Sustainable Power Generation and Supply (SUPERGEN 2012), 04 April 2013.
- Bansilal Bairwa; Mrinal Sarvagya; Sanjeeva Kumar; Arpita Banik "Battery Swapping Mobile Service Station for Electric Vehicles Application" 2021 Second International Conference on Smart Technologies in Computing, Electrical and Electronics (ICSTCEE), 15 February 2022.
- Xian Zhang; Guibin Wang "Optimal dispatch of electric vehicle batteries between battery swapping stations and charging stations" 2016 IEEE Power and Energy Society General Meeting (PESGM), 14 November 2016.
- Sayantika Sikdar; Madhavi Damle "IoT Solutions for Electric Vehicles

